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MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

V SEMESTER B.TECH. (ELECTRICAL & ELECTRONICS ENGINEERING)

END SEMESTER EXAMINATIONS, DECEMBER 2024

POWER SYSTEM ANALYSIS [ELE 3121]

REVISED CREDIT SYSTEM

Time: 3 Hours **19 November 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Two generators operate in parallel supplying a motor through a transmission line having transformers at both the ends. There are lagging loads at the generator and motor terminal buses. (3)

i) Draw a single line diagram of the sample power system described above.

ii) Draw reactance diagrams of the system for a load flow study and fault analysis, justifying the assumptions made.

1B. A 20MVA, 11kV, three phase alternator has a reactance of 15% and supplies a static load over a parallel transmission line having transformers (T_1 and T_2) at both ends with the ratings as follows: (4)

Static load: 5MVA, 0.8pf lagging, 11kV

Transformer T_1 (Δ-Y): 25MVA, 12.5/132kV with leakage reactance of 10%

Transformer T_2 (Y-Δ): 20MVA, 132/11kV with leakage reactance of 10%

Reactance of each Line: j500Ω

Draw the reactance diagram with all reactances marked in per-unit and select a base of 20MVA, 132kV in the transmission line.

1C. For the system shown in Figure 1C, estimate the rating of circuit breaker 'C'. Consider 30 KV and 100 MVA as the base values in the generator circuits. (3)

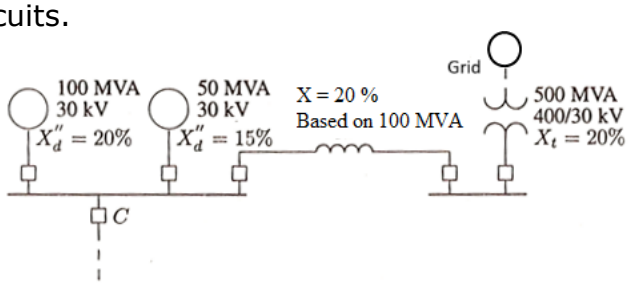


Figure 1C

2A. The single line diagram of a power system is as shown in Figure 2A. Bus 2 is loaded with 50 MVA, at 11 KV with load power factor of 0.8 lag. A three-phase symmetrical fault occurs at Bus 2. Estimate a) fault (4)

current and b) generator bus voltage using method of internal voltage. Consider 50 MVA and 11KV as the base values in the circuit.

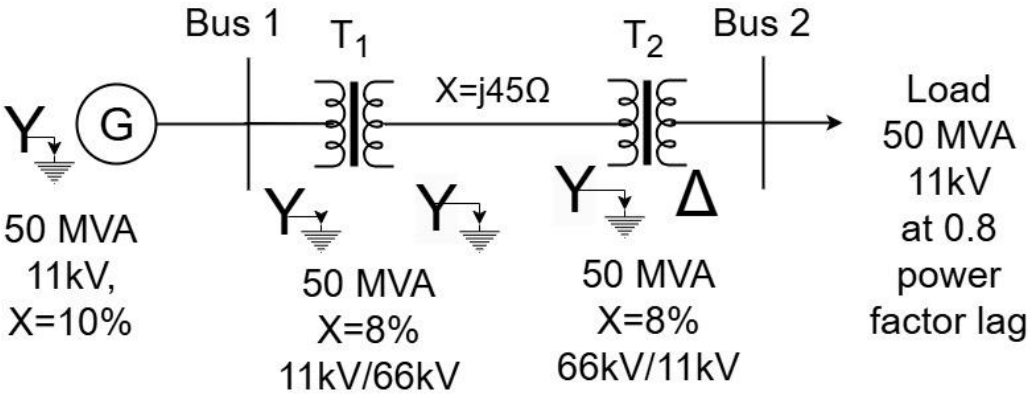


Figure 2A

2B. A 20 MVA, 11 kV synchronous generator has subtransient, transient and steady state reactances of 0.15 pu, 0.22 pu and 0.3 pu respectively based on its rating. For a symmetrical fault at its terminals on no load, determine 3

(i) Subtransient symmetrical fault currents in pu.
(ii) Transient and sustained symmetrical fault currents in pu.
(ii) Momentary current of circuit breaker in Amp.

2C. Two 6.6 kV, 3-phase 10 MVA alternators are connected to a common bus. Each alternator has X_1=0.15 p.u., X_2=0.1 pu, X_0=0.05 pu and x_n=0.06 pu. An unsymmetrical fault occurs on the bus. Calculate the fault current in kA, if the fault is i) bolted (or solid) L-G , ii) L-L through a fault reactance of 0.01 pu & iii) solid L-L-G fault. 3

3A. A single line diagram is shown in Fig 3A. The ratings are Machines 1 & 2: 100 MVA, 20 kV, X_1=X_2=0.2 pu, X_0=0.05 pu, X_n=0.05 pu 4
T1 & T2: 100 MVA, 20Δ/345Y kV, X=0.08 pu
Line: X_1=X_2=0.15 pu, X_0=0.5 pu
(i)Draw the sequence diagrams
(ii) Compute the fault current in kA if a double line to ground fault occurs at the midpoint of the line.



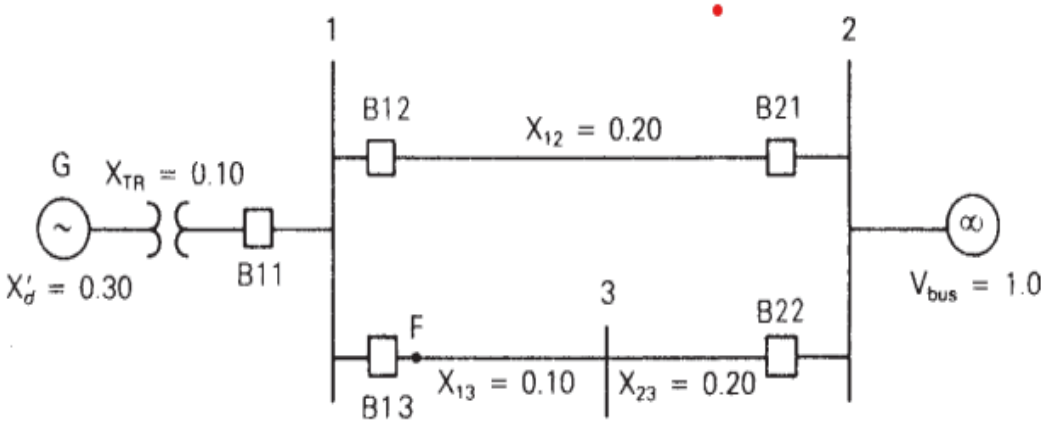
Figure 3A

3B. A 50 Hz generator supplies a real power of 0.77 pu to the infinite bus through a double circuit line and has H=2 MJ/MVA. The infinite bus voltage is 1∠0° and the magnitude of internal bus voltage E'=1.25 pu. A 3-phase short circuit fault occurs at the midpoint of the line such that the transfer reactance (X) is increased from 0.758 pu to 2 pu. The fault is cleared by isolating the line in 0.125 s. The post-fault transfer reactance is 0.95 pu. Comment on the transient stability of the system by numerically solving the swing equation for 0.35 s, taking a time interval of 0.05 s. 3

3C. Discuss the effect of the following components on the transient stability of the power system with proper justification. (i) Fast-acting excitation system, (ii) addition of series capacitor to the existing transmission 3

line, (iii) addition of auto-reclosure unit in circuit breaker and (iv) adding a parallel line to the existing line.

4A. A 50 Hz, Synchronous generator has H=5 MJ/MVA, and X_d'=0.3 p.u. The input mechanical power is 1 pu. The voltage behind transient reactance of generator is 1.15 p.u. Infinite bus voltage (V_{bus}) is 1+j0 p.u. A 3-phase fault occurs at F as shown in Fig 4A. Three cycles later, the fault extinguishes by itself. Due to a relay maloperation, all the circuit breakers remain closed. Use equal area criterion to compute the maximum generator power angle δ and comment on the transient stability of the system. 3



4B. A two-pole, 50 MVA, 11 kV generator is supplying full load at 0.8 power factor lagging. If the inertia constant of the moving parts of the generator is 6 MJ/MVA, calculate the energy stored when the generator is running at a synchronous speed of 3000 RPM. If the net input to the generator is suddenly increased to 62000 HP, calculate the acceleration produced. 3

4C. $$Ybus = j\begin{bmatrix} -7 & 3 & 4 \\ 3 & -8 & 5 \\ 4 & 5 & -9 \end{bmatrix}$$ 4

Table 4C: Bus Data

Bus No.	Bus Type	Generation (p.u.)		Load (p.u.)		Bus Voltage	
		P_G	Q_G	P_D	Q_D	V	δ
1	Slack	-	-	-	-	1.02	0
2	PQ	0.25	0.15	0.5	0.25	-	-
3	PQ	0	0	0.6	0.3	-	-

By using Gauss seidel method, evaluate the voltages of buses 1, 2 and 3 at the end of 1st iteration using the data given in Ybus matrix and Table 4C. Assume the acceleration factor as 1.6.

5A. Build the Ybus matrix for the given power system network. 3
The branch impedances of the lines are as follows:
Line 1-2: (10+j40) Ω; Line 1-4: (15+j50) Ω
Line 2-3: (5+j25) Ω; Line 2-4: (15+j20) Ω
Line 3-4: (10+j30) Ω
Assume that an impedance of (20+j40) Ω is connected between node 4 and the ground.

5B. Describe the step-by-step process of Gauss-Seidel method when PV buses are also included in the network along with PQ buses. 3

5C. $$Ybus = \begin{bmatrix} 20 - j50 & -10 + j20 & -10 + j30 \\ -10 + j20 & 26 - j52 & -16 + j32 \\ -10 + j30 & -16 + j32 & 26 - j62 \end{bmatrix}$$ 4

Table 5C: Bus Data

Bus No.	Bus Type	Generation (p.u.)		Load (p.u.)		Bus Volatge	
		P_G	Q_G	P_D	Q_D	V	δ
1	Slack	-	-	-	-	1.05	0
2	PQ	-	-	4	2.5	-	-
3	PV	2	-	0	-	1.04	-

Calculate the change in active & reactive power values at bus 2 $(i.e, \Delta P, \Delta Q)$ for the power system network presented in Ybus matrix and Table 5C.